In the earlier classes, we initiated the study of coordinate geometry where we studied about coordinate axes, coordinate plane, plotting of points in a plane, distance between two points, section formulae, etc. All these concepts form the basics of coordinate geometry. Let us now recall some of the basic formulae.

Recall

Distance between two points

Distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$|AB| = d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}\ .$$

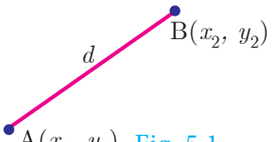


Fig. 5.1

Mid-point of line segment

The mid-point M, of the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

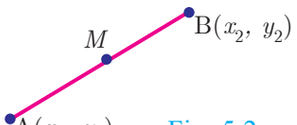


Fig. 5.2

Section Formula

Internal Division

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two distinct points such that point $P(x, y)$ divides AB internally in the ratio $m:n$.

Then the coordinates of P are given by $\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}\right)$.

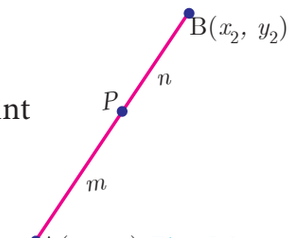


Fig. 5.3

External Division

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two distinct points such that the point $P(x, y)$ divides AB externally in the ratio $m:n$.

Then the coordinates of P are given by $\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}\right)$.

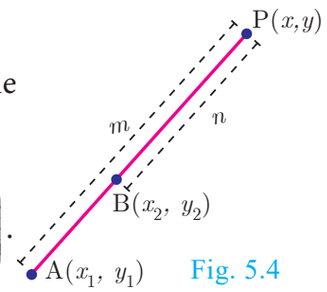


Fig. 5.4

Centroid of a triangle

The coordinates of the centroid G of a triangle with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are given by $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.

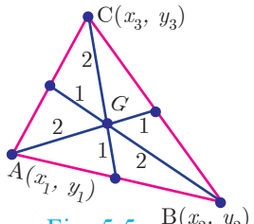


Fig. 5.5

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